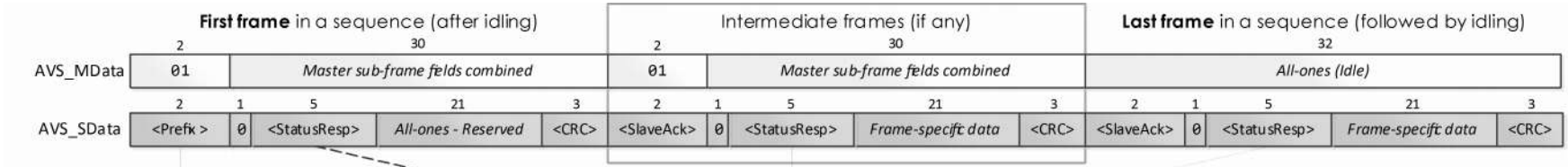
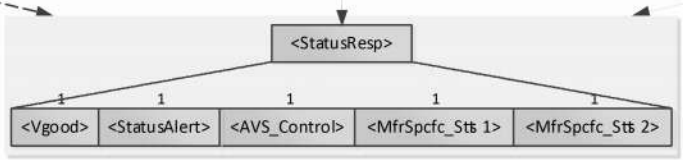
AVSBus Quick Guide

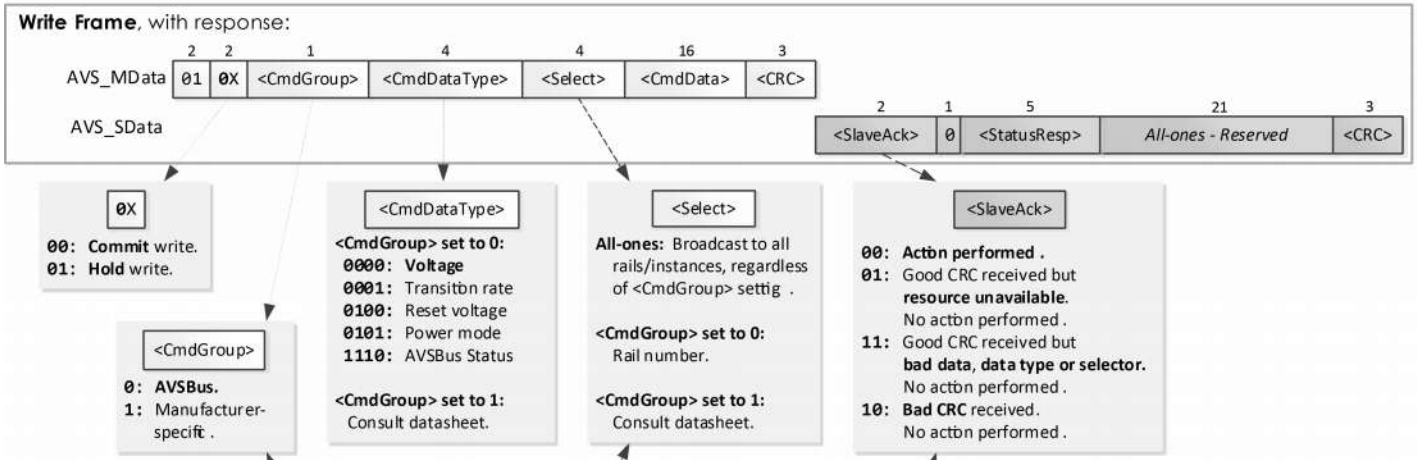
Provided by Texas Instruments, Inc.

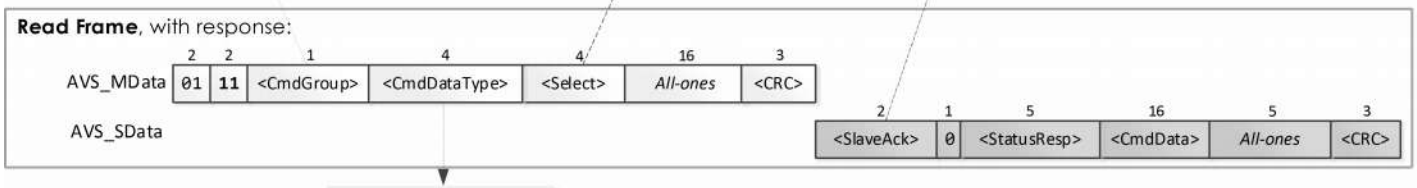
Ver. 1.0.1, 09/2015

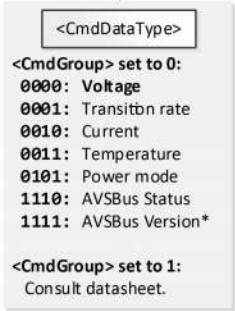












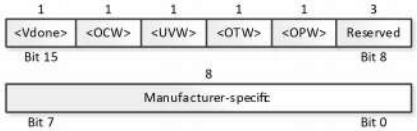
* AVSBus Version can only be used with <Select> set to Broadcast (All-ones).

Key Concepts

Data Launch: On rising edge of clock.
Data Capture: On falling edge of clock.

<CmdData> formats:
- Voltage: 16-bit, unsigned. LSB: 1 mV
- Trans. Rate: 8-bit Rise rate (top byte), 8-bit Fall rate (bottom byte), unsigned. LSB: 1 mV/µS
- Reset Voltage: 16-bit, all-zeroes required.
- Current: 16-bit, unsigned. LSB: 10 mA
- Temperature: 16-bit, signed. LSB: 0.1 °C
- Power Mode: 3-bit, encoded, LSB-aligned.
- AVSBus status: 16-bit, encoded.
- AVSBus version: 4-bit, encoded, LSB-aligned.

Data encodings:
- Power Mode: 000: Maximum efficiency
  001: Reserved
  010: Reserved
  011: Maximum power
  1XX: Manufacturer-specific .
- AVSBus status:



- AVSBus version: 0000: PMBus 1.3
  Others: Reserved.

Idle bus:
- AVS_MData held high.
- AVS_SData held high, but set low when slave issues an **interrupt** to the master.
- AVS_Clock held low (if stopped.)

Slave resynchronization : 34 consecutive ones received on AVS_MData.

CRC polynomial: CRC(x) = x^0 + x^1 + x^3